Spotify Genre Analysis

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Research Questions

- What are the least and the most popular genre? What‘s the best central tendency measure to represent the reality of the dataset?
- Is popularity influenced on a specific variable?
- What are the most different and the most similar couples of genre?

The Spotify Dataset

Column	Type	
Track id	Categorical	The Spotify ID for the track
artists	Categorical	The artists' names who performed the track
album_name	Categorical	The album name in which the track appears
Track name	Categorical	Name of the track
popularity	Numerical	calculated by algorithm
Duration	Numerical	The track length in milliseconds
explicit	Categorical	Whether or not the track has explicit lyrics
danceability	Numerical	how suitable a track is for dancing
energy	Numerical	measure of intensity and activity
key	Categorical	Measure of the pitches in a song
loudness	Numerical	Overall loudness of the track in decibels
mode	Categorical	Modality of the track
speechiness	Numerical	Presence of words in the track
acousticness	Numerical	Confidence measure of whether the track is acoustic
instrumentalness	Numerical	Predicts whether a track contains no vocals
liveness	Numerical	Detects the presence of an audience in the record
valence	Numerical	Musical positiveness of the track
tempo	Numerical	Beats per minute
Time signature	Numerical	How many beats in each bar
Track genre	Categorical	The genre of the track

Excluded from analysis

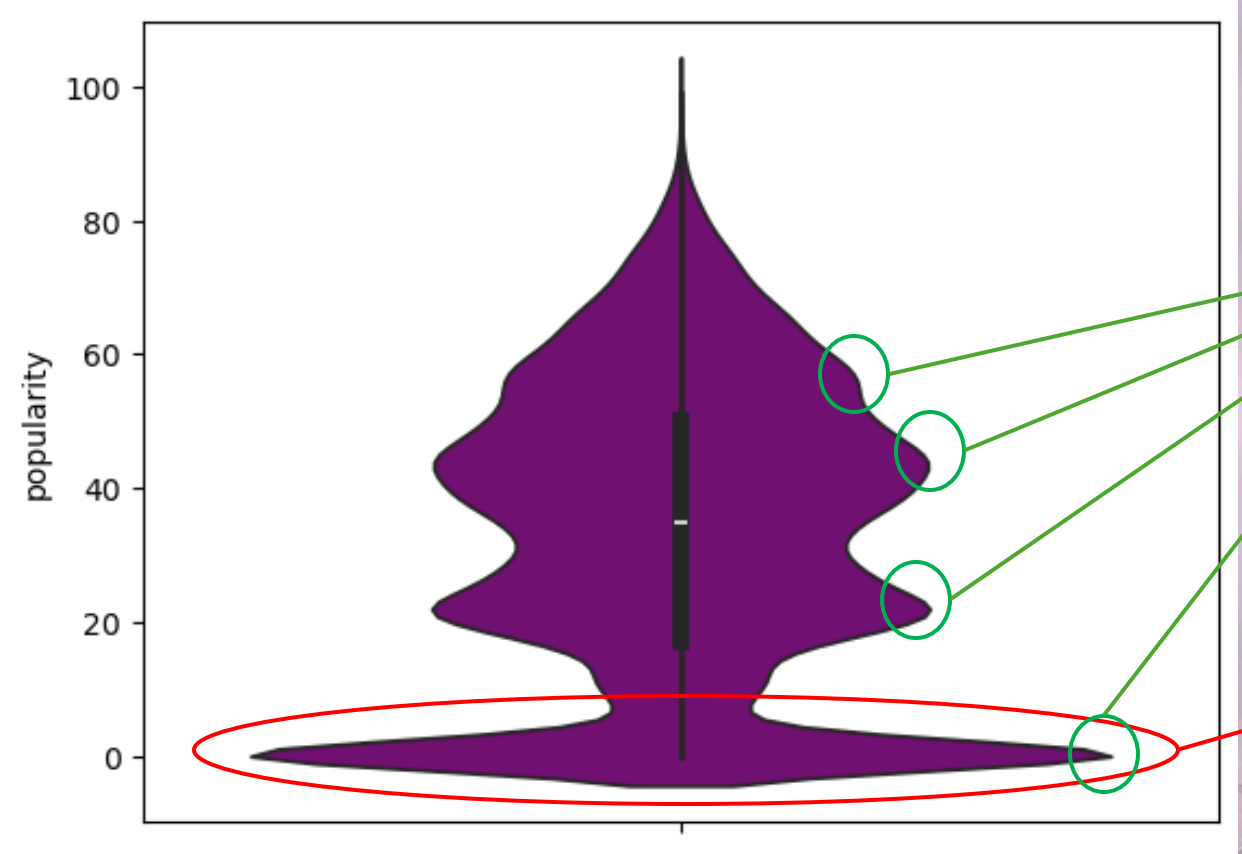
Included in analysis

Aggregation level

N° songs: 114000
N° Genres: 114
(1000 songs per genre)

Categorical variables were treated as **numerical** where appropriate, as they were encoded as binary (0–1) or ordinal values (e.g. -1 to 5)

The Popularity Distribution



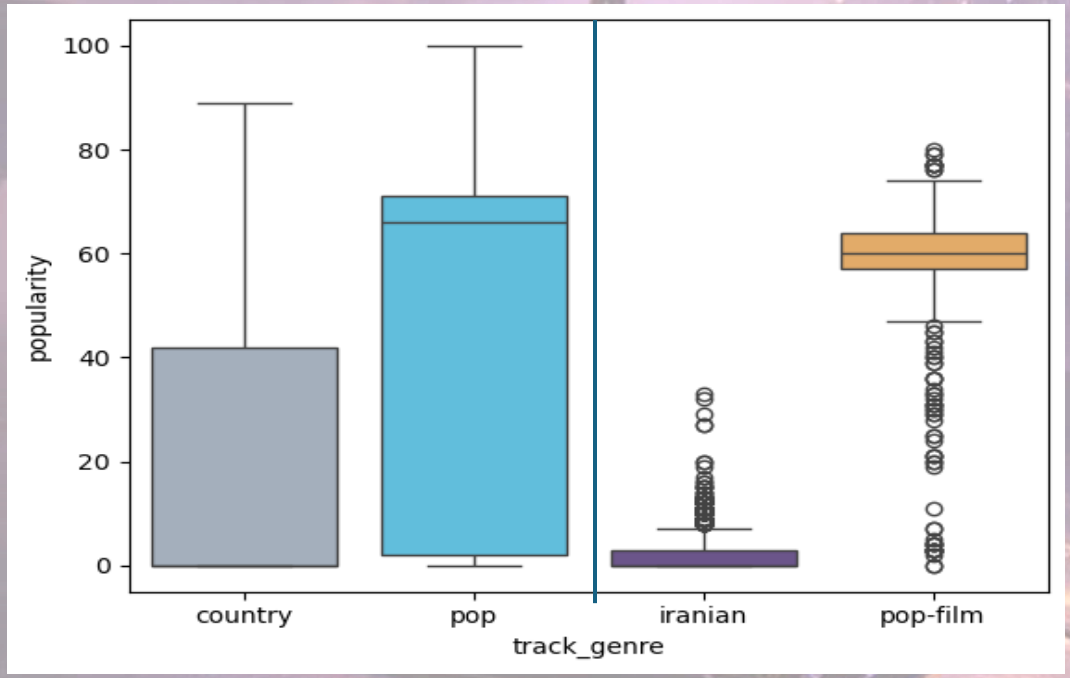
Multimodal Structure: there are different tiers of popularity

song popularity highly skewed toward low values

Median or Average?

Least popular genres by MEDIAN	
Track genre	Median popularity
country	0.0
iranian	0.0
jazz	0.0
latin	0.0
rock	0.0
romance	0.0
soul	0.0
Most popular genre by MEDIAN	
Track genre	Median popularity
pop	60

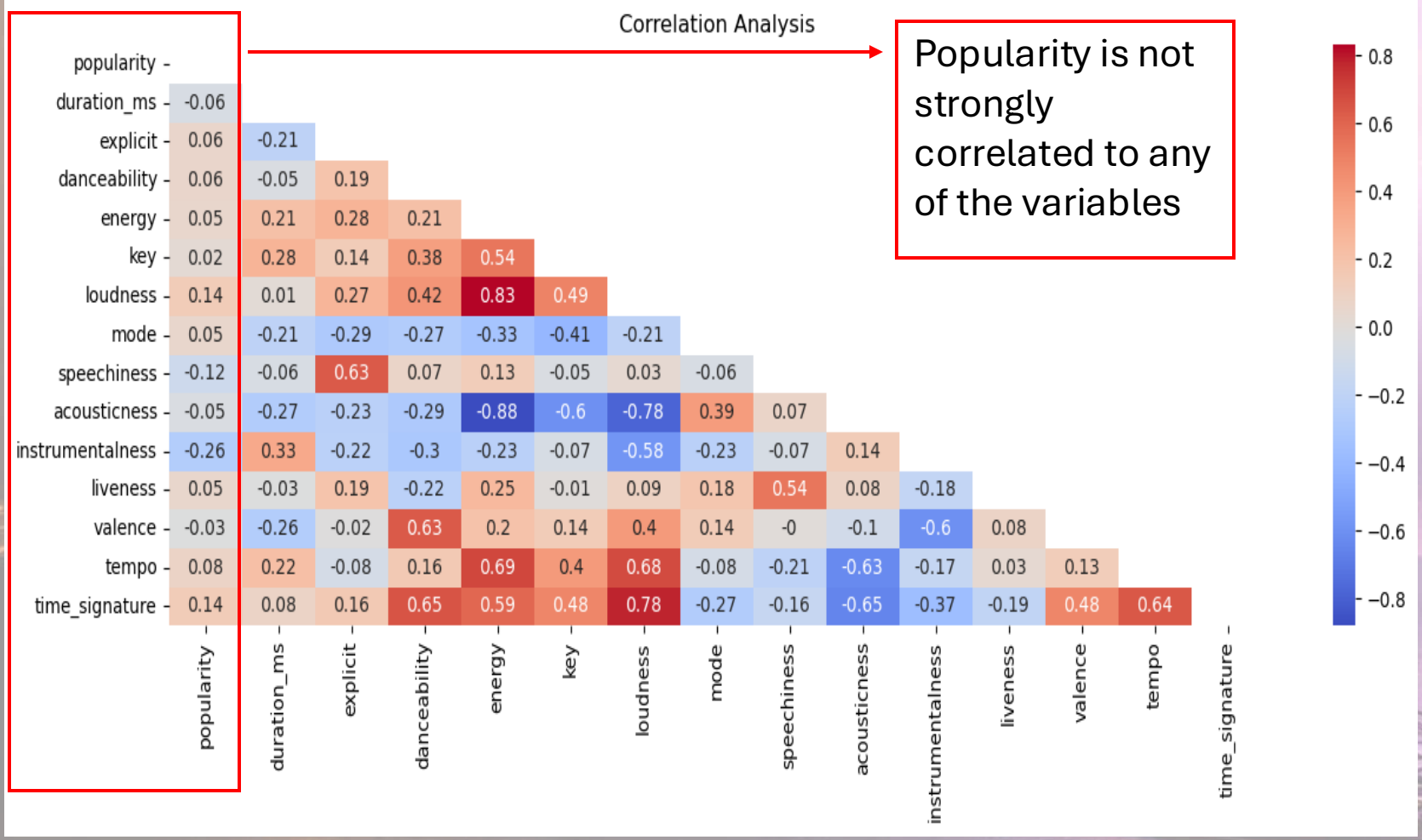
Least popular genre by MEAN		Most popular genre by MEAN	
Track genre	AVG popularity	Track genre	AVG popularity
iranian	2.21	Pop-film	59.28



Iranian VS Pop-film

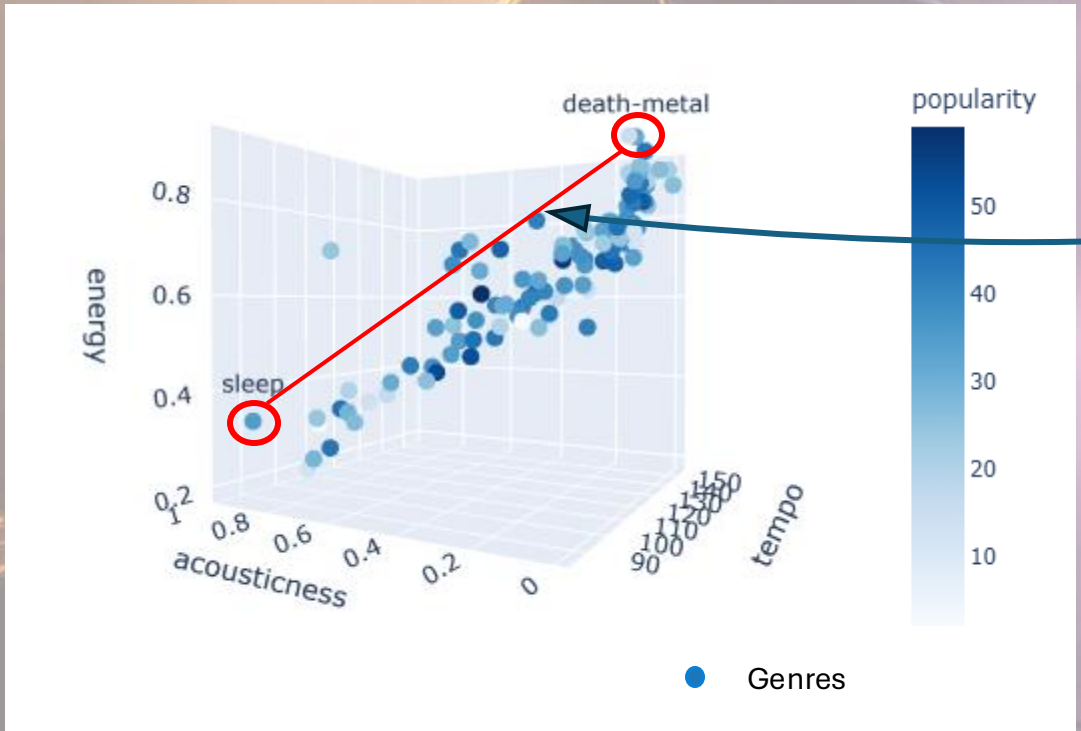
Variable	Tendency measure	Genre Iranian	Genre pop-film	Genre difference (%)
popularity	AVG	2.210	59.283	**2582.49**
Duration ms	AVG	319709.537	279657.084	-12.53
explicit	AVG	0.080	0.001	**-98.75**
danceability	AVG	0.301	0.597	**98.34**
energy	AVG	0.546	0.605	10.81
key	AVG	5.135	5.296	3.14
loudness	MEDIAN	-11.811	-7.586	-35.77
mode	AVG	0.547	0.557	1.83
speechiness	MEDIAN	0.057	0.041	-28.07
acousticness	MEDIAN	0.282	0.430	**52.48**
instrumentalness	AVG	0.590	0.008	**-98.64**
liveness	AVG	0.192	0.166	-13.54
valence	AVG	0.154	0.529	**243.51**
tempo	AVG	114.769	117.257	2.17
Time signature	AVG	3.728	3.899	4.59

Is popularity influenced by a specific variable?



Popularity is not strongly correlated to any of the variables

Distance Analysis – Idea



$$\sqrt{(\Delta\text{energy})^2 + (\Delta\text{acousticness})^2 + (\Delta\text{tempo})^2}$$

Which genres are the most distant from each other?
How "similar" or "dissimilar" are two genres?

Distance Analysis – Challenges

Observation: Some variables differ a lot in scale

	min	25%	50%	75%	max
duration_ms	8586.00	174066.00	212906.00	261506.00	5237295.00
explicit	0.00	0.00	0.00	0.00	1.00
danceability	0.00	0.46	0.58	0.70	0.98
energy	0.00	0.47	0.68	0.85	1.00
key	0.00	2.00	5.00	8.00	11.00
loudness	-49.53	-10.01	-7.00	-5.00	4.53

We want equal weights for all columns!

Solution: Scale all columns to [0,1]:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Distance Analysis – Result I

track_genre	iranian	romance	latin	detroit-techno	chicago-house	classical	jazz	...	grunge	sad	chill	k-pop	pop-film
track_genre													
iranian	0.00	0.51	0.53	0.42	0.46	0.49	0.52		0.58	0.66	0.62	0.62	0.61
romance	0.51	0.00	0.55	0.74	0.71	0.40	0.31		0.69	0.57	0.52	0.65	0.60
latin	0.53	0.55	0.00	0.49	0.41	0.74	0.52		0.41	0.48	0.49	0.42	0.48
detroit-techno	0.42	0.74	0.49	0.00	0.13	0.74	0.75		0.59	0.71	0.68	0.59	0.62
chicago-house	0.46	0.71	0.41	0.13	0.00	0.76	0.72		0.53	0.67	0.65	0.53	0.56
classical	0.49	0.40	0.74	0.74	0.76	0.00	0.40		0.79	0.66	0.58	0.76	0.69
jazz	0.52	0.31	0.52	0.75	0.72	0.40	0.00		0.59	0.50	0.44	0.57	0.52
⋮													⋮
indian	0.55	0.52	0.45	0.63	0.58	0.59	0.39		0.30	0.30	0.23	0.21	0.15
grunge	0.58	0.69	0.41	0.59	0.53	0.79	0.59		0.00	0.40	0.39	0.20	0.29
sad	0.66	0.57	0.48	0.71	0.67	0.66	0.50		0.40	0.00	0.15	0.34	0.35
chill	0.62	0.52	0.49	0.68	0.65	0.58	0.44		0.39	0.15	0.00	0.31	0.29
k-pop	0.62	0.65	0.42	0.59	0.53	0.76	0.57		0.20	0.34	0.31	0.00	0.14
pop-film	0.61	0.60	0.48	0.62	0.56	0.69	0.52	...	0.29	0.35	0.29	0.14	0.00

114 x 114 "Dissimilarity" Matrix

Normalized Euklidian Distance in 15 "Dimensions"

Distance Analysis – Result II

Most distant:

track_genre	track_genre	Distance
happy	sleep	1.00
drum-and-bass	sleep	1.00
chicago-house	sleep	0.99
hardstyle	sleep	0.98
party	sleep	0.98
j-idol	sleep	0.97
breakbeat	sleep	0.97
reggae	sleep	0.97
reggaeton	sleep	0.96

Most similar:

track_genre	track_genre	Distance
singer-songwriter	songwriter	0.00
reggaeton	latino	0.03
punk-rock	punk	0.04
indie	indie-pop	0.04
reggae	reggaeton	0.05
spanish	j-pop	0.06
j-rock	hard-rock	0.06
reggae	latino	0.06
house	electro	0.07

Thank you for listening to our ramblings!

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